ARMAN BABAEI

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PROFESSIONAL SUMMARY

Software engineer with professional C++ and Python experience in network software and test automation, alongside production Go development and reliability engineering. Research at Sharif, EPFL and ISTA spans programmable networking, experimental systems and performance analysis.

SKILLS

Programming	C++, Python, Go, C, Bash; additional experience with Rust, Kotlin and Java
Systems	Linux networking, concurrency, distributed systems, P4, QUIC, QEMU, Mininet
Testing and delivery	Git, automated testing, packet simulation, Docker, CI/CD, GitLab CI
Platform	Kubernetes, Helm, Flagger, Istio, GitOps, Prometheus, Grafana
Firmware projects	UEFI, EDK II, ACPI inspection and network-interface discovery

WORK EXPERIENCE

Tapsi September 2024 – Present
Proficient Platform Engineer *Tehran, Iran*

- **Platform Engineer (Apr'26 – Present):** Develop deployment automation and safer Kubernetes delivery workflows with Helm, Flagger, Istio and GitOps, including canary releases, rollback safety and observability.
- **SRE (Sep'25 – Apr'26):** Investigated production incidents and debugged failures across microservices, service-mesh traffic and deployment infrastructure; supported on-call operations.
- **Backend Engineer (Sep'24 – Sep'25):** Built passenger–driver matching features and rewrote Node.js services in Go, including services handling approximately 1,000 requests/second; maintained Go/Python/Node.js services using Kafka and Kubernetes.

Undisclosed Hedge Fund September 2023 – February 2024
Full-stack Developer Intern *Remote*

- Developed a Python/Django service for scheduling internal jobs, including REST APIs and frontend integration.

Mahsan Co. September 2020 – July 2022
C++/Python Developer *Tehran, Iran*

- Developed parallel network-relaying software in C++ that connected to ISP switches.
- Led development of a Python domain-specific language for end-to-end test automation, using simulated network packets to exercise network behavior.
- Served as Scrum Master for a team of five.

EDUCATION

Sharif University of Technology September 2024 – Present
M.S. in Computer Engineering (Networks), pending defense
Thesis approved; oral defense pending scheduling. GPA: 18.50/20.

Sharif University of Technology September 2019 – July 2024
B.S. in Computer Engineering
GPA: 19.48/20; rank: 15/190. English: TOEFL 113/120.

AVAILABILITY

Available from November 2026, subject to visa and work authorization for Belgium. Employer support required.

RESEARCH EXPERIENCE

Sharif University of Technology August 2025 – Present
L4S and Volumetric Video Streaming *Tehran, Iran*

- Implemented a DualPI2 active queue management prototype in P4/BMv2 and compared its behavior with a Linux reference implementation for low-latency networking.
- Built a reproducible experiment harness with C traffic fixtures and Python automation for QEMU and Mininet. Combined packet captures, transport metrics and queue counters to validate congestion behavior and analyze results.
- Evaluated delivery and rendered quality for 3D Gaussian Splatting streaming. In the studied setup with competing classic traffic, measured 99th-percentile bottleneck queueing delay of approximately 2.3 ms with L4S versus 49.2 ms in the classic baseline. Experiment code

Stony Brook University October 2024 – Present
Adaptive Leader Election for PBFT *Remote*

- Develop a modular Go implementation separating consensus, view changes and leader election; investigate adaptive selection using Byzantine-robust aggregation of node progress signals. Implementation

EPFL DEDIS July 2023 – September 2023
Fairness in Decentralized Storage *Lausanne, Switzerland*

- Developed a mathematical model for fairness in Swarm, extended a Go simulator with churn and heterogeneous node behavior, and designed and evaluated a network-shuffling mechanism to improve fairness.

ISTA SPiDerS July 2022 – February 2023
Parallel Execution in Blockchain Protocols *Austria*

- Reproduced prior results, built an experimental test bench and analyzed blockchain workloads; contributed implementation work for parallel smart-contract execution.

SELECTED SOFTWARE PROJECTS

UEFI System Information Viewer
C, EDK II, QEMU

- Contributed ACPI table inspection, network-interface discovery through UEFI protocols and display fixes to an interactive firmware-level system information viewer.

Touch
Kotlin, Go, Android, Wear OS

- Developed standalone phone and watch applications integrating haptic playback, voice communication, persistent message delivery and a Go backend, with automated tests and release workflows.

SELECTED PUBLICATIONS

CHIRON: Accelerating Node Synchronization without Security Trade-offs in Distributed Ledgers. arXiv, revised 2026.

Tit-for-Token: Understanding Fairness when Forwarding Data by Incentivized Peers in Decentralized Storage Networks. arXiv, 2023.

TEACHING AND RECOGNITION

Algorithm design teacher (2019–2020); university teaching assistant experience includes Operating Systems and Data Structures and Algorithms. Bronze medal, Iran National Olympiad in Informatics (2018).